



Niche dynamics and climate change sensitivity in western Palearctic *Zamenis* ratsnakes (Reptilia: Colubridae)

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Abstract. Climate change poses a significant threat to ectothermic, dispersal-limited reptiles such as *Zamenis* snakes in the western Palearctic. This study used ensemble ecological models to assess the potential distribution of six *Zamenis* species under climate change scenarios (SSP1-2.6 and SSP5-8.5). In addition, we examined niche dynamics through overlap, equivalency, and similarity analyses. Results reveal distinct ecological niches for each species, primarily influenced by temperature and precipitation. Projected distribution shifts vary across species, with some species like *Z. persicus* and *Z. situla* showing potential for range expansion, while others such as *Z. lineatus* and *Z. scalaris* face significant habitat loss. *Z. hohenackeri* and *Z. longissimus* exhibit complex patterns of both expansion and contraction. Niche analyses indicate varying degrees of overlap among species, with a trend toward niche conservatism. This study underscores the importance of niche ecology in understanding climate change impacts and highlights the need for species-specific conservation strategies.

Keywords: allopatric distribution, climate change, conservation, ecological niche modeling, niche conservatism, niche divergence, western Palearctic.

Introduction

Species on Earth occupy unique ecological niches shaped by geological processes such as continental drift and glaciations (Pelegrin et al., 2021). Within these niches, competition and selection pressures drive adaptation and evolution (Angert and Schemske, 2005; Zhao, Zhang and Wei, 2019). This interplay between species and niches influences phenotypic changes, functional diversity, and geographic distribution (Sexton et al., 2009), ultimately shaping biodiversity patterns. Understanding this relationship is crucial for elucidating speciation events and the processes that generate biodiversity. By examining species-environment interactions, we can gain

insights into past and future evolutionary trajectories (McKinnon et al., 2004; Soberon and Peterson, 2005).

Ecological patterns, both spatial and temporal, play a significant role in shaping speciation processes (Wiens, 2011). Niche conservatism, characterized by the preservation of fundamental ecological requirements between closely related taxa, has been shown to drive speciation even during periods of environmental change (Wiens and Graham, 2005). However, niche divergence, the evolution of divergent ecological niches, can also contribute to speciation dynamics. This is particularly evident in functional lineage diversification, where different

species within a phylogenetic group occupy distinct ecological niches (Ford et al., 2016). The concept of niche, which encompasses a species' environmental requirements, interactions with its surroundings, and resilience to environmental change, is a fundamental ecological concept that underpins speciation (Poisot et al., 2011). By differentiating species from one another, niche divergence ensures that each species experiences unique biotic and abiotic processes within its environment. This phenomenon, in turn, promotes the adaptation of species to diverse ecological conditions, enabling them to persist and evolve (Pyron et al., 2014).

Ecological niche modeling (ENM) has emerged as a powerful tool for understanding species distributions and environmental preferences (Zhu et al., 2013). This technique has broad applications in ecology, evolution, biogeography, and conservation biology (Phillips, Anderson and Schapire, 2006; Peterson et al., 2011; Gherghel, 2021). Recent studies have employed ENM to investigate species niche dynamics, revealing insights into habitat suitability, biogeography, speciation and niche evolution (Ahmadi et al., 2018; Hua, Cardillo and Bromham, 2019; Warren et al., 2019; Vaissi, 2022; Vaissi et al., 2023; Khoshnamvand et al., 2024). However, a critical challenge in ENM is the potential for spurious niche divergence signals arising from spatial autocorrelation in environmental variables. It can artificially inflate the observed niche overlap between species, leading to misleading conclusions about their ecological relationships. To address this issue and ensure the validity of ENM results, the use of null models is recommended (Sillero et al., 2021). Null models represent a statistical approach that permits testing whether the observed species' overlap in environmental space may be confidently attributed to differing environmental tolerances, or spatial autocorrelation of environmental factors not directly related to the species' niche (Di Cola et al., 2017; Broennimann et al., 2023).

Reptiles, especially snakes, are particularly vulnerable to climate change due to their ectothermic nature and temperature-dependent life processes (Lourenço-de-Moraes et al., 2019; Martinez et al., 2024). Climate change disrupts their habitat, body temperature regulation, and food sources, leading to population declines and potential extinction (Winter et al., 2016; Doherty et al., 2020). This study focuses on the genus *Zamenis* (Reptilia, Squamata, Colubridae), which encompasses six snake species distributed widely across the western Palearctic region (Salvi et al., 2018). This group of snakes is of great interest for biogeographic research, providing insights into species distribution patterns and the impacts of environmental change. A previous study by Salvi et al. (2018) provided a detailed analysis of the genus *Zamenis*, including its phylogenetic relationships, taxonomic status, and biogeographic distribution. In this study, the authors incorporated the species previously known as *Rhinechis scalaris* into the genus *Zamenis*, establishing *Zamenis* and *Rhinechis* as a single genus. This taxonomic revision clarifies their close evolutionary relationship and prompts reevaluations of their conservation status and management strategies. Overall, it underscores the importance of accurate taxonomy in guiding conservation efforts for these snakes. Ara et al. (2022) conducted a specific study on the distribution modeling of *Zamenis persicus*, an endemic species to Iran. Using ENM techniques, the authors projected the potential distribution of this species under current and future climate scenarios. Their findings indicated that climate change may pose a threat to *Z. persicus* by shrinking the areas suitable for its persistence (Ara et al., 2022).

Despite numerous investigations into the evolutionary, systematic, and biogeographic dimensions of this species group (e.g., Schatti and Utiger, 2001; Schatti and Monsch, 2004; Araújo, Thuiller and Pearson, 2006; Musilová, Zava-

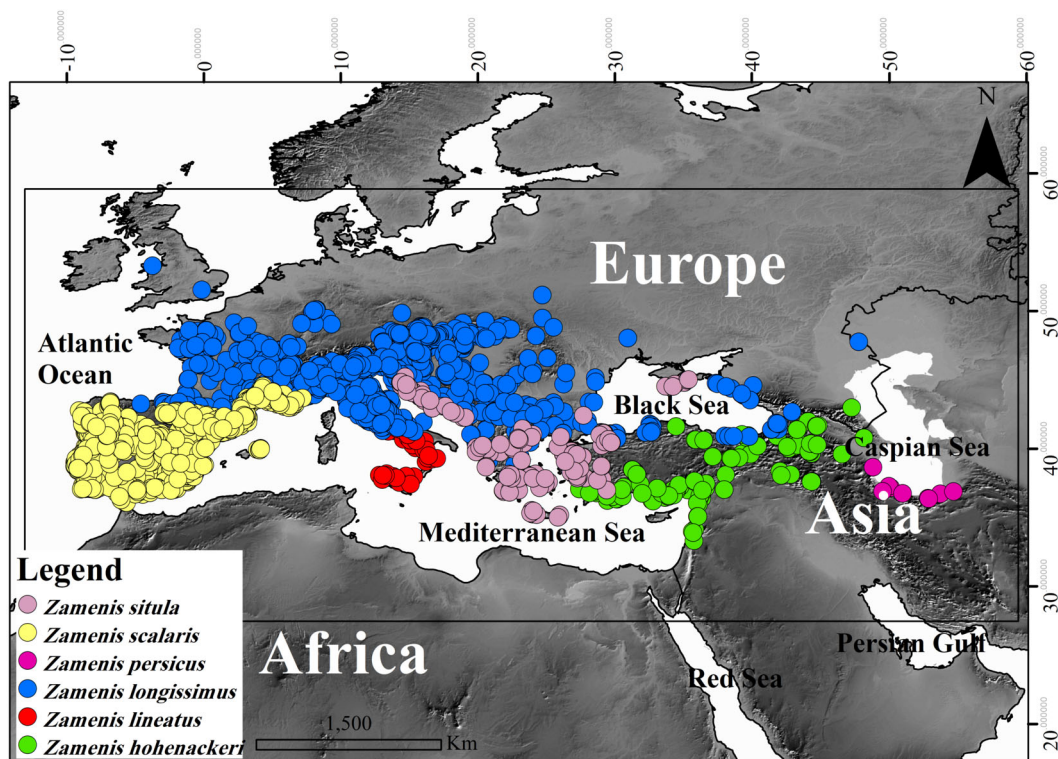


Figure 1. Study area. The occurrence records of six species of *Zamenis* with different colours are shown on the map. The background illustrates the topography of the study area.

dil and Kotlík, 2007; Carvalho et al., 2010; Afrasiab and Mohamad, 2011; Lelièvre et al., 2012; Salvi et al., 2018), there is a conspicuous gap on a comprehensive evaluation of their potential distributional shifts under climate change scenarios. To address this gap, this research employs ENM techniques to determine whether the genus *Zamenis* has maintained a consistent niche throughout its evolutionary history or if niche divergence and/or conservatism have shaped its trajectory. Additionally, we explore the potential future distributions (2081-2100) of *Zamenis* species under various climate change scenarios, aiming to provide valuable insights into the anticipated impacts of climate change on these snakes and to inform conservation strategies for their protection.

Materials and methods

Study area and input data

The study area spans 10° West to 60° East in Longitude and 31° to 59° North in Latitude, encompassing diverse habitats across western Asia, the Levantine Region, and continental Europe, including the Iberian Peninsula and Great Britain (fig. 1). This region encompasses the entire range of *Zamenis* species, providing a comprehensive representation of their ecological niche preferences. Occurrence records for *Zamenis* species were obtained from various sources, including museum databases, herpetological publications, and online databases. Detailed information about these sources is provided in supplementary table S1. To ensure the quality and accuracy of the data, it underwent a two-step data cleaning and enhancement process. First, the georeferenced data were meticulously examined for errors and inconsistencies in geographic coordinates. This was done using appropriate data validation techniques to ensure the reliability of the occurrence points (Chapman, 2005). Additionally, for records sourced from online platforms, we prioritized data with high accuracy ratings and detailed observation descriptions. Second, to minimize geographic

Table 1. Schoener’s D index and *p*-values for the equivalency (E) and similarity (S) test for niche divergence (D) and conservatism (C). The correlation circle illustrates the changing relevance between two species ranges along the first two principal axes.

Species	Schoener’s D	E (D)	S (D)	E (C)	S (C)	Correlation circle (%)		Figure reference
						PC1	PC2	
<i>Zamenis hohenackeri</i> vs <i>Z. lineatus</i>	0.06	0.001	1	1	0.003	39.06	33.93	fig. 3A
<i>Zamenis hohenackeri</i> vs <i>Z. longissimus</i>	0.33	0.18	1	0.80	0.003	34.48	22.93	fig. 3B
<i>Zamenis hohenackeri</i> vs <i>Z. persicus</i>	0.04	0.32	1	0.82	0.003	46.56	27.86	fig. 3C
<i>Zamenis hohenackeri</i> vs <i>Z. scalaris</i>	0.43	1	1	0.001	0.003	36.11	27.97	fig. 3D
<i>Zamenis hohenackeri</i> vs <i>Z. situla</i>	0.54	0.92	1	0.31	0.001	40.32	31.89	fig. 3E
<i>Zamenis lineatus</i> vs <i>Z. longissimus</i>	0.13	0.52	0.65	0.41	0.27	38.16	21.81	fig. 3F
<i>Zamenis lineatus</i> vs <i>Z. persicus</i>	0.08	0.27	1	0.50	0.002	43.69	27.36	fig. 3G
<i>Zamenis lineatus</i> vs <i>Z. scalaris</i>	0.32	0.75	0.81	0.19	0.13	37.56	29.96	fig. 3H
<i>Zamenis lineatus</i> vs <i>Z. situla</i>	0.38	0.01	0.67	0.98	0.33	36.34	30.54	fig. 3I
<i>Zamenis longissimus</i> vs <i>Z. persicus</i>	0.38	1	1	0.25	0.33	38.48	20.29	fig. 3J
<i>Zamenis longissimus</i> vs <i>Z. scalaris</i>	0.35	1	0.97	0.001	0.02	45.08	23.28	fig. 3K
<i>Zamenis longissimus</i> vs <i>Z. situla</i>	0.29	1	1	0.001	0.002	36.75	21.45	fig. 3L
<i>Zamenis persicus</i> vs <i>Z. scalaris</i>	0.11	0.57	0.89	0.38	0.13	39.38	29.42	fig. 3M
<i>Zamenis persicus</i> vs <i>Z. situla</i>	0.09	0.10	0.79	0.87	0.18	38.53	30.35	fig. 3N
<i>Zamenis scalaris</i> vs <i>Z. situla</i>	0.59	1	1	0.001	0.001	36.58	29.22	fig. 3O

sampling biases and enhance the interpretation of habitat suitability analyses and niche overlap tests, the R package *spThin* (Aiello-Lammens et al., 2015) was employed to spatially rarefy the occurrence records for each *Zamenis* species. For this, one locality was retained for every 1 km to ensure a balance between data density and spatial representation. The number of occurrence records for each *Zamenis* species before and after the cleaning process is provided in supplementary table S2. Additionally, figure 1 illustrates the geographic distribution of the studied species, including their occurrence points.

Bioclimatic variables

To accurately assess the niche consistency of *Zamenis* species and project their potential distributions during recent (1970 to 2000) and future (2081 to 2100) climate scenarios, we employed a comprehensive set of bioclimatic variables. These variables represent crucial climatic conditions that significantly influence the distribution and abundance of *Zamenis* snakes. For recent and future bioclimatic variables, we sourced data from the WorldClim dataset (v. 2.1) (Fick and Hijmans, 2017), which provides global climate data at a resolution of 30 arc-seconds (~1 km²). To project the potential future distributions of *Zamenis* species under different climate change scenarios, we considered two future climate projections representing low and high emissions pathways. These projections aligned with the Shared Socioeconomic Pathways (SSPs) 1-2.6 and 5-8.5, respectively. The SSP 1-2.6 scenario assumes a moderately lower emissions pathway, resulting in an estimated global temperature rise of approximately 1.6°C by the end of the century. In contrast, the SSP 5-8.5 scenario represents a higher emissions pathway, potentially leading to a global temperature increase of up to 4.6°C by the end of the century (Eyring et al., 2016).

For the future climate projections, we obtained data from the Coupled Model Intercomparison Project Phase 6

(CMIP6), the latest generation of global climate models (Eyring et al., 2016). To ensure a comprehensive assessment of future climate conditions, we selected two global climate models (GCMs) for each projection scenario: the Goddard Institute for Space Studies’ Earth System Model (GISS-E2-1-G) and the Meteorological Research Institute Earth System Model (MRI-ESM2-0) (Sun et al., 2022). These models were chosen based on their demonstrated performance, ranking among the top performers for Eurasia among 32 models (Sun et al., 2022). The ensemble average of these GCMs was utilized as the final future climate data, providing a balanced representation of potential climate scenarios for our projections.

We conducted a Pearson correlation analysis to identify variables exhibiting high multicollinearity. This step was crucial to reduce redundancy and ensure the independent contribution of each variable to the ecological niche modeling (ENM) models. The threshold for removing correlated variables was set at $r \geq |0.8|$, a commonly accepted threshold in ecological research (Borcard et al., 2018). Finally, eleven bioclimatic variables were selected based on their relevance to the ecology of *Zamenis* species: Mean Diurnal Range (Bio2), Isothermality (Bio3), Temperature Seasonality (Bio4), Temperature Annual Range (Bio7), Mean Temperature of Wettest Quarter (Bio8), Mean Temperature of Warmest Quarter (Bio10), Mean Temperature of Coldest Quarter (Bio11), Precipitation Seasonality (Bio15), Precipitation of Driest Quarter (Bio17), Precipitation of Warmest Quarter (Bio18), and Precipitation of Coldest Quarter (Bio19).

Species distribution modeling

The species distribution modeling in this study utilized an ensemble of nine algorithms implemented through the “biomod2” package (v 4.2-4) (Thuiller et al., 2023)

in the R programming language (v. 4.2.0). These algorithms were selected based on their established performance in ecological research, following default settings recommended by Guisan, Thuiller, and Zimmermann (2018). The chosen algorithms include: Artificial Neural Networks (ANN, CV.ann = 5, 5 Fold Cross Validation + 3 Repetitions); Classification Tree Analysis (CTA, CV.tree = 50, 5 Fold Cross-Validation); Flexible Discriminant Analysis (FDA); Generalized Additive Models (GAM, algo = 'GAM_mgcv'); Generalized Boosted Models (GBM, n.trees = 1000, 3 Fold Cross-Validation); Generalized Linear Models (GLM, type = 'quadratic', interaction.level = 1, stepwise procedure using Akaike Information Criterion (AIC)); Maximum Entropy (MaxEnt, maximum iterations = 500, https://biodiversityinformatics.amnh.org/open_source/maxent/); Multivariate Adaptive Regression splines (MARS, simple with no interaction); and Random Forest (RF, n.trees = 1000).

For each *Zamenis* species, a unique set of 1000 pseudo-absences was generated to ensure species-specific environmental variability was adequately captured. Pseudo-absences were selected using a background selection method based on environmental covariates to minimize bias in the model training process (Wisz and Guisan, 2009; Barbet-Massin et al., 2012; Guisan, Thuiller and Zimmermann, 2017). To evaluate the performance of the models and mitigate overfitting, a 10-fold cross-validation strategy with blocking was employed. This method, implemented using the blockCV package in the R environment, ensured that geographical autocorrelation was accounted for during model validation (Valavi et al., 2018). For each fold of the cross-validation, the trained models were used to predict the presence-absence of the target species in the test data. Predictions from all models were aggregated, and the average prediction for each data point was calculated (Valavi et al., 2018).

Model evaluation was conducted using Cohen's kappa (KAPPA), Area Under Curve (AUC), and True Skill Statistics (TSS) as default metrics (Cohen, 1960; Hanley and McNeil, 1982; Fielding and Bell, 1997; Allouche, Tsoar and Kadmon, 2006). A TSS or KAPPA value between 0.60 and 0.90 indicates fair to good model performance, with a value of 1 indicating perfect agreement (Allouche, Tsoar and Kadmon, 2006). AUC values of 0.90 and above are considered good, 0.60 to 0.90 are considered medium, and 0.60 and below are considered poor (Phillips, Anderson, and Schapire, 2006). Variable importance was assessed using permutation importance and partial dependence plots, in addition to the mean variable importance scores across the different algorithms. This approach provides a more nuanced understanding of the contribution of each variable to the model predictions, considering interactions and non-linear relationships among predictors (Guisan, Thuiller and Zimmermann, 2017).

The "BIOMOD EnsembleModeling()" function was employed to generate ensemble models. Two ensembling options, committee average and weighted mean, were evaluated to reduce the number of outputs. The ensembling process involved mixing all models, algorithms, pseudo-absence sampling, and cross-validation runs. TSS was used

as the evaluation standard for forming committees and determining weights. Models with a TSS greater than or equal to 0.8 were retained in the final ensemble. This rigorous approach ensures the reliability of the ensemble models in predicting species distribution under varying conditions (Guisan, Thuiller and Zimmermann, 2017). *Z. hohenackeri* had 82 out of 90 replicates with $TSS \geq 0.8$, *Z. lineatus* had 88, *Z. longissimus* had 86, *Z. persicus* had 83, *Z. scalaris* had 89, and *Z. situla* had 88 (supplementary table S3). Finally, we compared projected climatic suitability derived from ensemble modeling with the observed distribution of the target species. By overlaying model predictions onto known occurrence data, we visually assessed areas where projected suitability extended beyond the documented range of species occurrences.

Species range change

To assess and visualize potential species range change (SRC) of *Zamenis* species under climate change scenarios, we utilized the BIOMOD_RangeSize function within the biomod2 package. This function generates spatial maps illustrating areas where species may experience gains or losses in suitable conditions, alongside summary tables outlining range change metrics. These metrics include Loss (number of pixels predicted to become unsuitable), Absent (number of pixels predicted to remain unsuitable), Stable (number of pixels predicted to remain suitable), and Gain (number of pixels predicted to become suitable). It's important to note that these estimates are based on model predictions and do not directly reflect the species' actual area of occupancy, as they are derived from binarized model predictions using a specified threshold (Guisan, Thuiller and Zimmermann, 2017).

In addition to absolute measures, three relative metrics were derived to further elucidate the potential impacts of climate change on *Zamenis* species distributions:

1. Percentage Loss: This metric represents the proportion of currently occupied sites that are predicted to be lost. It is calculated as $(Loss / (Loss + Stable))$.
2. Percentage Gain: This metric represents the proportion of new sites that the species is predicted to gain, relative to the species' current distribution size. It is calculated as $(Gain / (Loss + Stable))$.
3. Range Change: This metric indicates the net change in the species' range size, considering both gains and losses.

It is calculated as $(Percentage\ Gain - Percentage\ Loss)$. These metrics offer valuable insights into how climate change may impact the distribution of *Zamenis* species. Analyzing these metrics across various climatic scenarios allows for a comprehensive understanding of potential range shifts for each species (Guisan, Thuiller and Zimmermann, 2017).

Niche analysis

To assess the ecological niche overlap between each pair of *Zamenis* species, niche analysis was conducted independently for each species pair using the R package ecospat

(Di Cola et al., 2017). A PCA-env (principal component analysis-environment) framework and an ordination method were employed in the analysis. The climate variables from the study area were transformed into two-dimensional spaces. The two-dimensional environmental space was then projected onto a 100×100 grid of cells bounded by the minimum and maximum PCA values of the background data. Smoothed occurrence probabilities of each species were estimated using a kernel density function for each species in each grid cell. To quantify the degree of niche overlap between species, Schoener's D metric was utilized, which ranges from 0 (indicating no overlap) to 1 (indicating complete overlap) (Broennimann et al., 2012).

To determine whether niche overlap was statistically significant (Schoener's D), equivalency and similarity tests were conducted. The niche equivalency test assesses whether two realized niches estimated from occurrence data are equivalent by randomly permuting occurrences between ranges. The niche similarity test evaluates whether the niches of two species are more or less similar than expected by chance by randomly shifting niches within accessible conditions in the study region (Broennimann et al., 2012; Di Cola et al., 2017; Warren et al., 2021). For the niche similarity test, the "accessible conditions" were defined by the entire range of environmental variation captured by the climate data from the study area (excluding presence data for *Zamenis* species). This encompasses the full spectrum of potentially suitable conditions within the region. We examined niche conservatism (alternative = greater, i.e. the niche overlap is more equivalent/similar than random) and niche divergence (alternative = lesser, i.e. the niche overlap is less equivalent/similar than random) using the alternative option in *ecospat* (Broennimann et al., 2012; Di Cola et al., 2017). To verify each hypothesis, 1000 permutations were performed. R (v. 4.2.0) was utilized for all analyses (www.project.org).

Results

Species distribution modeling

The ensemble models developed for *Zamenis* species demonstrated high-quality performance, with TSS, AUC, and KAPPA scores ranging from 0.75 to 1 (supplementary table S4). Bio-climatic variables representing variability in temperature and precipitation were analyzed for six species of *Zamenis* snakes (supplementary table S5). The potential distribution of six *Zamenis* species under recent (1970-2000), and future climate projections (2081-2100) for SSP1-2.6 and SSP5-8.5 scenarios is depicted in figs. 2A-F and 3. Below are details for each species.

Zamenis hohenackeri

The TSS values ranged from 0.84 to 0.97, with RF and MaxEnt models performing best. AUC values ranged from 0.92 to 1, with the highest values in GLM, RF, FDA, and MARS models. KAPPA values ranged from 0.75 to 0.93, with RF and MaxEnt models being the highest (supplementary table S4). The most important variables were BIO15 at 15.47%, BIO19 at 14.00%, and BIO18 at 13.13% (supplementary table S5). The future projections for *Z. hohenackeri* indicate substantial shifts in suitable habitat by 2081-2100. Compared to its current range, the species is projected to expand, especially in Europe, while contracting at the southern edge of its distribution. It is expected to retain some suitable habitats in western Asia and the Middle East (figs. 2 and 3). Under the SSP1-2.6 scenario, habitat loss is projected at 12.60%, with a 33.94% gain in suitable habitats, resulting in a SRC of 21.33%. In the SSP5-8.5 scenario, habitat loss is anticipated to be 32.63%, while the gain is expected to be 69.62%, leading to a SRC of 36.98%.

Zamenis lineatus

TSS values ranged from 0.91 to 1, with GBM, RF, CTA, ANN, and FDA models achieving the highest scores. AUC values were consistently 1 across all models. KAPPA values ranged from 0.85 to 1, with GBM, RF, ANN, and FDA models performing best (supplementary table S4). The key variables were BIO7 at 18.22%, BIO18 at 16.01%, and BIO4 at 10.69% (supplementary table S5). The projections for *Z. lineatus* in 2081-2100 indicate a contraction in suitable habitat. Under the SSP1-2.6 scenario, the contraction is expected to be relatively modest, with a 19.71% loss of habitat and a 23.34% gain, leading to a SRC of 3.63%. This scenario suggests that suitable habitats may decrease primarily in southern Italy, while there could be potential increases in western Europe (figs. 2 and 3). Conversely, the SSP5-8.5 scenario predicts a more pronounced contraction, with a 42.72%

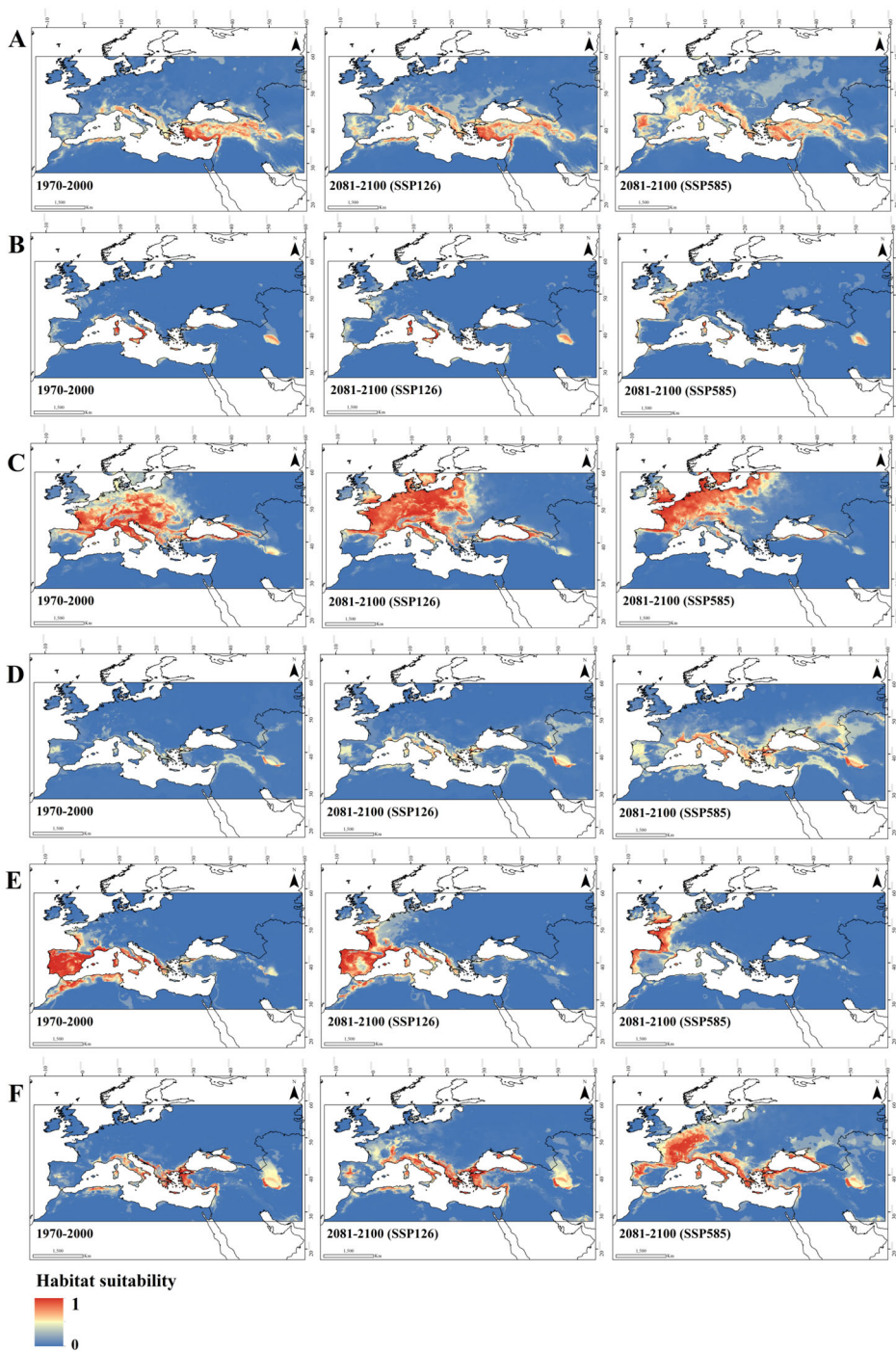


Figure 2. Recent (1970-2000) and future (2081-2100) climatic suitability for six *Zamenis* species based on the consensus model under the optimistic (SSP1-2.6) and pessimistic (SSP5-8.5) scenarios. A: *Zamenis hohenackeri*, B: *Zamenis lineatus*, C: *Zamenis longissimus*, D: *Zamenis persicus*, E: *Zamenis scalaris*, F: *Zamenis situla*.

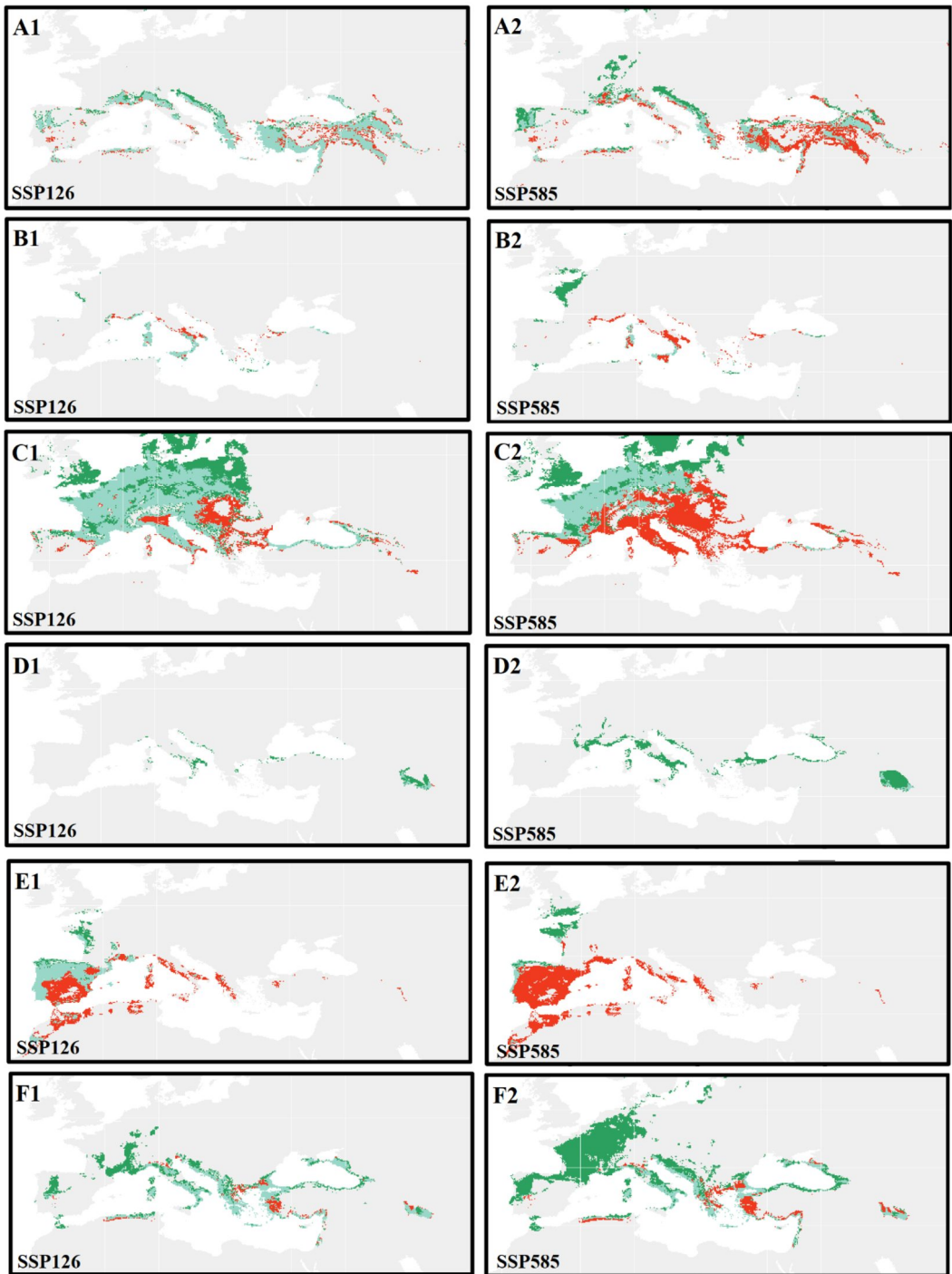


Figure 3. Species range change of *Zamenis* species in recently suitable habitats (gain/loss) by 2081-2100 under optimistic (SSP1-2.6) and pessimistic (SSP5-8.5) scenarios. A: *Zamenis hohenackeri*, B: *Zamenis lineatus*, C: *Zamenis longissimus*, D: *Zamenis persicus*, E: *Zamenis scalaris*, F: *Zamenis situla*. The color scheme used in the maps is as follows: red represents areas indicating loss, green denotes gain, blue signifies stability, and gray indicates absence.

habitat loss and a 69.88% gain, resulting in a SRC of 27.16%. This scenario indicates a substantial reduction in suitable habitats, despite the potential for considerable habitat gain.

Zamenis longissimus

TSS values ranged from 0.93 to 0.99, with the highest value for MaxEnt. AUC values were consistently 1 across all models. KAPPA values ranged from 0.90 to 0.97, with MARS and MaxEnt models being the highest (supplementary table S4). The highest importance was seen in BIO17 at 13.82%, BIO11 at 13.73%, and BIO2 at 13.30% (supplementary table S5). The future projections for *Z. longissimus* indicate a significant northward expansion of suitable habitat, potentially extending into northern and western Europe, especially under the pessimistic scenario. This expansion includes potential gains across these regions. Meanwhile, a substantial portion of the distribution range along the southern edges is expected to be lost (figs. 2 and 3). Under the SSP1-2.6 scenario, a 27.02% habitat loss and a 73.31% gain are projected, resulting in a SRC of 46.29%. Conversely, the SSP5-8.5 scenario forecasts a 63.14% habitat loss and a 68.95% gain, leading to a much smaller SRC of 5.80%.

Zamenis persicus

TSS values ranged from 0.84 to 1, with RF and FDA models performing best. AUC values ranged from 0.90 to 1, with the highest values in GBM, RF, FDA, and MARS models. KAPPA values ranged from 0.75 to 0.95, with FDA performing best (supplementary table S4). The most important variables were BIO17 at 16.94%, BIO19 at 14.20%, and BIO15 at 13.64% (supplementary table S5). For *Z. persicus*, recent habitat suitability trends show a concentration of its range within the Hyrcanian forests. Future projections suggest that the species' distribution will expand into the border countries of the Mediterranean and northern Turkey while continuing to occupy the

Hyrcanian forests (figs. 2 and 3). Under the SSP1-2.6 scenario, a 1.01% habitat loss and a 248.95% gain are predicted, resulting in a SRC of 247.93%. The SSP5-8.5 scenario forecasts an even more dramatic shift, with a 0.74% habitat loss and an 813.07% gain, leading to a SRC of 812.33%.

Zamenis scalaris

TSS values ranged from 0.95 to 1, with GLM, GBM, RF, CTA, ANN, and FDA models achieving the highest scores. AUC values were consistently 1 across all models except MaxEnt (0.99). KAPPA values ranged from 0.96 to 1, with GLM, GBM, RF, ANN, and FDA models performing best (supplementary table S4). The key variables were BIO4 at 20.40%, BIO11 at 15.91%, and BIO10 at 11.87% (supplementary table S5). *Z. scalaris* is predicted to be highly vulnerable to climate change, with substantial future habitat changes. In recent climate conditions, the species was sparsely distributed across southwestern Europe. Projections suggest significant habitat loss, particularly on the southern edge of its distribution range (figs. 2 and 3). Under the SSP1-2.6 scenario, there is a 34.62% loss of suitable habitat and a 20.44% gain, resulting in a SRC of -14.18%. The SSP5-8.5 scenario predicts a more severe impact, with an 87.73% habitat loss and a 24.21% gain, leading to a SRC of -63.52%.

Zamenis situla

TSS values ranged from 0.89 to 1, with GBM, RF, GAM, ANN, FDA, and MARS models achieving the highest scores. AUC values were consistently 1 across all models. KAPPA values ranged from 0.83 to 1, with ANN performing best (supplementary table S4). The most important variables were BIO10 at 17.11%, BIO4 at 12.46%, and BIO11 at 11.55% (supplementary table S5). Recent habitat suitability for *Z. situla* includes Mediterranean border countries of southern Europe, western Asia, north Africa, and the Hyrcanian forests. Future

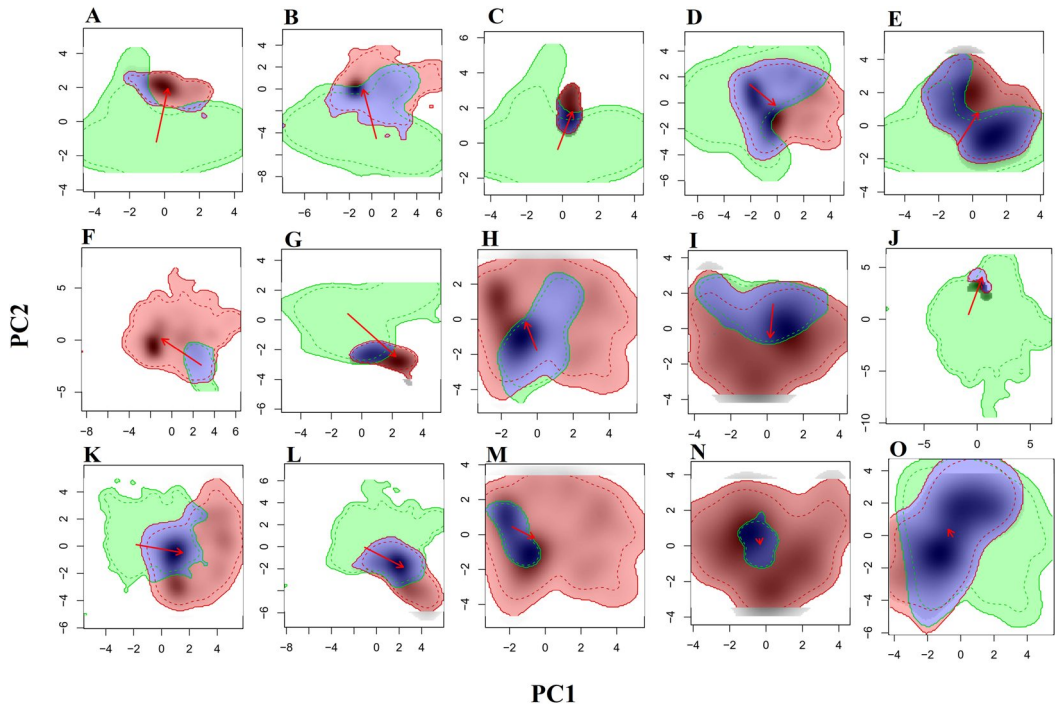


Figure 4. Panels (A-O) illustrate the niche overlap values between two species distribution ranges (see table 1), along the two first axes of the PCA. The solid and dashed lines represent 100% and 75% of the available (background) environment, respectively. The overlap is measured using Schoener's D metric. A: *Zamenis hohenackeri* vs *Z. lineatus*, B: *Zamenis hohenackeri* vs *Z. longissimus*, C: *Zamenis hohenackeri* vs *Z. persicus*, D: *Zamenis hohenackeri* vs *Z. scalaris*, E: *Zamenis hohenackeri* vs *Z. situla*, F: *Zamenis lineatus* vs *Z. longissimus*, G: *Zamenis lineatus* vs *Z. persicus*, H: *Zamenis lineatus* vs *Z. scalaris*, I: *Zamenis lineatus* vs *Z. situla*, J: *Zamenis longissimus* vs *Z. persicus*, K: *Zamenis longissimus* vs *Z. scalaris*, L: *Zamenis longissimus* vs *Z. situla*, M: *Zamenis persicus* vs *Z. scalaris*, N: *Zamenis persicus* vs *Z. situla*, O: *Zamenis scalaris* vs *Z. Situla*. The colors in figure represent the following: Green: This color represents the first species in the comparison. Red: This color represents the second species in the comparison. Blue: This color represents the overlap between the two species.

projections, especially under the SSP5-8.5 scenario, anticipate a substantial increase in suitable habitat in western and southern Europe, while retaining some areas within its current range, similar to the SSP1-2.6 scenario. However, there is also expected to be habitat loss, particularly at the southern edge of its distribution range (figs. 2 and 3). Under the SSP1-2.6 scenario, there is an 11.12% loss of suitable habitat and a 73.81% gain, resulting in a SRC of 62.68%. The SSP5-8.5 scenario predicts a more dramatic impact, with a 26.38% habitat loss and a 239.38% gain, leading to a SRC of 212.99%.

Niche analysis

The correlation circle explained by the first two principal components in the PCA-env analysis ranged from 58.20% to 74.42% across all possible pairwise comparisons of *Zamenis* species (table 1). The assessment of paired niche overlap, using the D metric, revealed a spectrum ranging from low ($D = 0.04-0.13$) to moderate ($D = 0.29-0.38$), and finally to substantial overlap ($D = 0.43-0.59$) (table 1, fig. 4). The niche equivalency test statistically supported the hypothesis of niche divergence ($p < 0.05$) in only 2 out of the 15 pairwise comparisons

across *Zamenis* species (table 1). For example, the comparison between *Z. hohenackeri* and *Z. lineatus* showed a low Schoener's D (0.06) with a significant p -value for niche equivalency ($E(D) = 0.001$). Conversely, the hypothesis of niche conservatism found support in 4 paired comparisons (table 1). For instance, the comparison between *Z. hohenackeri* and *Z. scalaris* exhibited moderate niche overlap with Schoener's D (0.43), and significant p -values for both equivalency ($E(C) = 0.001$) and similarity ($S(C) = 0.003$). Significant values were observed for 9 paired comparisons related to conservatism hypotheses in the niche similarity test (table 1). An example is the comparison between *Z. hohenackeri* and *Z. longissimus*, with moderate Schoener's D (0.33), indicating moderate niche overlap and significant p -values for similarity ($S(C) = 0.001$). Table 1 and fig. 4 illustrate these patterns, showing varying degrees of niche overlap and divergence among the *Zamenis* species.

Discussion

This study presents a comprehensive analysis of the ecological niche characteristics and potential future range dynamics of six *Zamenis* species, shedding light on their evolutionary history. These findings highlight the importance of developing targeted conservation strategies that address the specific vulnerabilities of each species, especially in the face of anticipated climate change impacts.

Snakes are a diverse and widespread group of reptiles that play an important role in ecosystem health. However, their spatial ecology has received less attention compared to other terrestrial vertebrates, with few studies focused on *Zamenis* species. This is likely due to the challenges associated with studying snakes, as they are often cryptic and difficult to observe (Boback et al., 2020). Despite these challenges, the limited studies on *Zamenis* species have provided some insights into their spatial ecology,

such as the seasonal movements of *Z. hohenackeri* and *Z. longissimus* (Capula et al., 2006; Ara et al., 2022), and the relatively sedentary nature of *Z. scalaris* (Martinez-Freiria, Lorenzo and Lizana, 2019). The limited migratory patterns observed in snakes are attributed to their ectothermic physiology (Southwood and Avens, 2010; McCue, Lillywhite and Beaupre, 2012), especially within *Zamenis* species (Salvi et al., 2017; Martinez-Freiria, Lorenzo and Lizana, 2019), and pose a challenge to the unlimited dispersion hypothesis. This hypothesis suggests that species can potentially disperse and colonize all suitable climatic areas under future climate change scenarios, without considering dispersal limitations such as barriers or rates of movement. However, this assumption may be unrealistic for snakes, whose dependence on external heat sources constrains their dispersal capabilities and adaptation to changing environmental conditions (Sahlean et al., 2014). Furthermore, in many regions, human-made barriers such as roads and highways further restrict snake dispersal, making the unlimited dispersion hypothesis less applicable. Therefore, this study highlights the importance of considering the limitations of snakes' spatial ecology and the realistic constraints on their dispersal when using climate change models to assess their range dynamics. This consideration is crucial for accurately understanding the potential impacts of climate change on the conservation of snake populations.

Climate change models can provide valuable insights into the potential impacts of climate change on snake range dynamics (Needleman, Neylan and Erickson, 2017; Piquet et al., 2021). These models can simulate how the distribution of suitable habitat is likely to change under different climate scenarios. By comparing the current distribution of snakes with the projected distribution of suitable habitats, we can identify areas where snakes are likely to experience range contractions or expansions. Up to now,

most studies on the use of climate change models for snakes have focused on examining seasonal movements and habitat use (e.g., Freedman et al., 2009; Croak et al., 2013; Pomara et al., 2014; Todd and Nowakowski, 2021; Subedi et al., 2022). However, there is a growing interest in using ENMs to examine niche-based interactions between snakes and other species (e.g., Pelegrin et al., 2021; Esparza-Estrada et al., 2023; Hidalgo-Licona, Flores-Villela and Velasco, 2023). This is important for understanding how climate change is likely to affect the distribution and abundance of snake populations. The findings of this study highlight the need for targeted conservation strategies for *Zamenis* species.

All *Zamenis* species exhibited varying degrees of dependence on bioclimatic variables, particularly temperature and precipitation, highlighting the crucial role of climatic factors in shaping their ecological niches. Species such as *Z. persicus* and *Z. situla* are projected to benefit from climate change with significant habitat gains, suggesting that these species might thrive if they can successfully track suitable habitats and manage potential losses. In contrast, *Z. scalaris* and *Z. lineatus* could face substantial range reductions if they fail to adapt to the changing conditions, underscoring the need for targeted conservation management of these populations. For *Z. hohenackeri* and *Z. longissimus*, the climate change projections reveal a more complex scenario, characterized by both potential range expansions and contractions. This variability highlights the need for nuanced conservation strategies that consider the specific ecological and evolutionary contexts of each species. Overall, our findings emphasize the importance of addressing species-specific responses to climate change, recognizing both opportunities and challenges under future scenarios. Effective conservation strategies should focus on habitat protection, enhancing connectivity, and implementing adaptive management practices to mitigate the impacts of climate

change and bolster species resilience (e.g., Littlefield et al., 2019).

From an evolutionary perspective, genetic studies have shown that *Z. hohenackeri* and *Z. situla* experienced range contractions during the last glacial period (LGM), with the Lebanon Mountains and Northern Anatolia being the likely refugia for these lineages (Salvi et al., 2018). Conversely, *Z. longissimus* displayed an expansion pattern after the LGM (Musilová et al., 2010). These findings suggest that climatic variables may have significantly influenced the distribution of these species in the region. Additionally, the Messinian Salinity Crisis, which occurred approximately 5.96 to 5.33 million years ago, could have been a major factor affecting their distribution. This crisis, which primarily impacted an area overlapping with the current range of *Zamenis* species, might have led to partial or complete desiccation in the region, contributing to historical distribution shifts (Orszag-Sperber, Rouchy and Blanc-Valleron, 2000). Thus, the present distribution patterns of *Zamenis* could be a result of allopatric speciation driven by vicariance, where physical barriers led to the separation and divergence of populations.

Climatic refugia have been crucial in shaping the distribution and evolutionary history of *Zamenis* snakes, providing essential havens during environmental upheavals like the ongoing global warming trend (Musilová et al., 2010). The Mediterranean Basin, especially its northern and eastern regions, exemplifies such a refugium, offering a diverse range of habitats – forests, grasslands, and rocky slopes – that support *Zamenis* species amidst environmental stress (Medail and Quezel, 1999; Cuttelod et al., 2009; Médail and Diadema, 2009; Valavanidis and Vlachogianni, 2011; Temunovic et al., 2013; Doxa, Kamarianakis and Mazaris, 2022). This region's varied topography, including mountains, valleys, and coastlines, provides a patchwork of microclimates beneficial to these species (Aschmann, 1973). Our study supports this by identifying similar stable climatic areas

within the Mediterranean Basin, underscoring the historical significance of these refugia in the persistence and speciation of *Zamenis* snakes.

Similarly, the Hyrcanian forest, with its diverse microclimates and proximity to the Caspian Sea, serves as a crucial refuge for *Z. persicus*. Historical evidence indicates that during glacial periods, this forest likely supported various species, including *Z. persicus*, which relies on its varied habitats for thermoregulation (Akhami et al., 2010; Naderi et al., 2014; Asadolahi, Salmanmahiny and Sakieh, 2017; Ahmadi et al., 2018; Erichsen et al., 2018; Amiri et al., 2021; Ramezani et al., 2023). Preserving this habitat is vital for the conservation of *Z. persicus* and similar species amidst climate change, highlighting the need for habitat restoration and species-specific research.

Recent studies on European and Mediterranean reptiles reveal a complex interaction between niche conservatism and divergence during diversification. For example, the evolution of European green lizards follows a pattern of allopatric speciation, niche conservatism, and gradual phenotypic change (Enriquez-Urzelai et al., 2022). In contrast, the *Lacerta trilineata* group exemplifies rapid radiation without strong niche conservatism, showcasing significant ecological diversification within complex landscapes (Ahmadzadeh et al., 2013). Similarly, the *Tarentola mauritanica* species complex underwent both niche divergence and conservatism during its diversification in the Miocene and Pleistocene, with patterns shaped by temperature and humidity variables (Rato et al., 2015). Vaissi et al. (2023) also documented niche dynamics in Snake-Eyed Skinks, finding both niche divergence and conservatism among *Ablepharus* species. Collectively, these studies underscore that closely related reptile species often inhabit distinct habitats and respond differently to climatic variables, highlighting the pivotal role of ecology in driving speciation processes.

In this study, we conducted niche analyses to evaluate the ecological niche overlap among six

Zamenis species. Our results revealed a spectrum of niche overlap, ranging from low to moderate to substantial, indicating diverse levels of ecological specialization within the genus. The niche equivalency and similarity tests further shed light on niche dynamics, with statistical support for niche divergence in only 2 out of 15 pairwise comparisons. This suggests that, while some species have developed distinct ecological niches, there is a broader pattern of niche conservatism within the *Zamenis* genus. Despite environmental changes, certain ecological preferences or constraints appear to persist, contributing to the overall stability of niche characteristics across these species. This trend of niche conservatism aligns with findings from other reptile groups (e.g., Morales-Castilla et al., 2010) and reinforces the idea that speciation in *Zamenis* may be influenced by the preservation of ancestral ecological traits.

Understanding the mechanisms driving niche evolution and the factors influencing range dynamics is essential for developing effective conservation strategies for *Zamenis* species, particularly in the context of a changing climate. Given the observed tendency towards niche conservatism and the varying degrees of ecological niche overlap among these species, it is crucial to address conservation efforts with a multifaceted approach (Wiens et al., 2010). This could involve developing cross-border conservation policies to manage and protect habitats that span multiple jurisdictions, ensuring that conservation actions are coordinated and comprehensive (Tingley, Darling and Wilcove, 2014). Additionally, species-specific conservation measures tailored to the unique ecological requirements of each *Zamenis* species could enhance the effectiveness of these strategies. By integrating these approaches, conservation efforts can be better aligned with the ecological needs of the species, thereby improving the chances of maintaining their populations and mitigating the impacts of environmental change (Likens and Lindenmayer, 2012).

While this study presents important insights into the ecological niches and range dynamics of *Zamenis* species, it is essential to recognize that the nature of the approach used is primarily correlative. Correlative models, such as ENMs, rely on observed species-environment relationships and are thus constrained by their assumptions and the quality of input data. These models do not fully account for factors like species' dispersal abilities, biotic interactions, or physiological constraints, which are crucial for understanding species' vulnerabilities to climate change (Kearney and Porter, 2009; Peterson, Papeş and Soberón, 2015). Future research could benefit from integrating mechanistic models that incorporate biological processes, such as energy budgets, thermal tolerances, and species' dispersal capacities, to provide a more detailed and biologically meaningful understanding of species' vulnerabilities to climate change (Leroux et al., 2013; Briscoe et al., 2023; Tourinho and Vale, 2023). These mechanistic models can complement correlative approaches by offering a more nuanced view of species' adaptive capacities and responses under various climate scenarios, leading to more robust conservation strategies (Leroux et al., 2013; Hill et al., 2015).

Conclusion

This study provides an in-depth analysis of the ecological niches and potential future range dynamics of six *Zamenis* snake species, providing valuable insights into their evolutionary history and conservation needs. By analyzing their dependence on bioclimatic factors and forecasting range shifts under climate change scenarios, we identify critical species-specific vulnerabilities and opportunities for tailored conservation strategies. The integration of climate change models with ecological niche analyses proves essential for understanding how species-specific interactions and habitat preferences drive their distribution and abundance. Our findings reveal

varying degrees of niche overlap among *Zamenis* species, highlighting different levels of ecological specialization. While some species exhibit niche divergence, a broader pattern of niche conservatism suggests that certain ecological traits are preserved despite environmental changes. This tendency towards niche conservatism underscores the importance of considering both historical and present-day factors when developing conservation strategies. To effectively conserve *Zamenis* species, a multifaceted approach is needed. This should include the development of cross-border conservation policies to ensure coordinated habitat management, along with species-specific conservation measures tailored to their unique ecological requirements. By aligning conservation efforts with the ecological needs of *Zamenis* species, we can enhance their resilience to climate change and improve the long-term viability of their populations.

Supplementary material. Supplementary material is available online at:
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